

Auteur: Shahin Jamal
Studierichting: Informatica
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$f(x, y, z) = xyz$ in $a(1, 1, 1)$ richting $h\left(\frac{1}{2}, 0, \frac{1}{2}\right)$ en $f(1, 1, 1) = 1$

$$Df(a, h) = \lim_{\lambda \rightarrow 0} \frac{f\left((1, 1, 1) + \lambda\left(\frac{1}{2}, 0, \frac{1}{2}\right)\right) - 1}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{\frac{1}{4}\lambda^2 + \lambda + 1 - 1}{\lambda}$$
$$= \lim_{\lambda \rightarrow 0} \left(\frac{1}{4}\lambda + 1\right) = 1$$

References